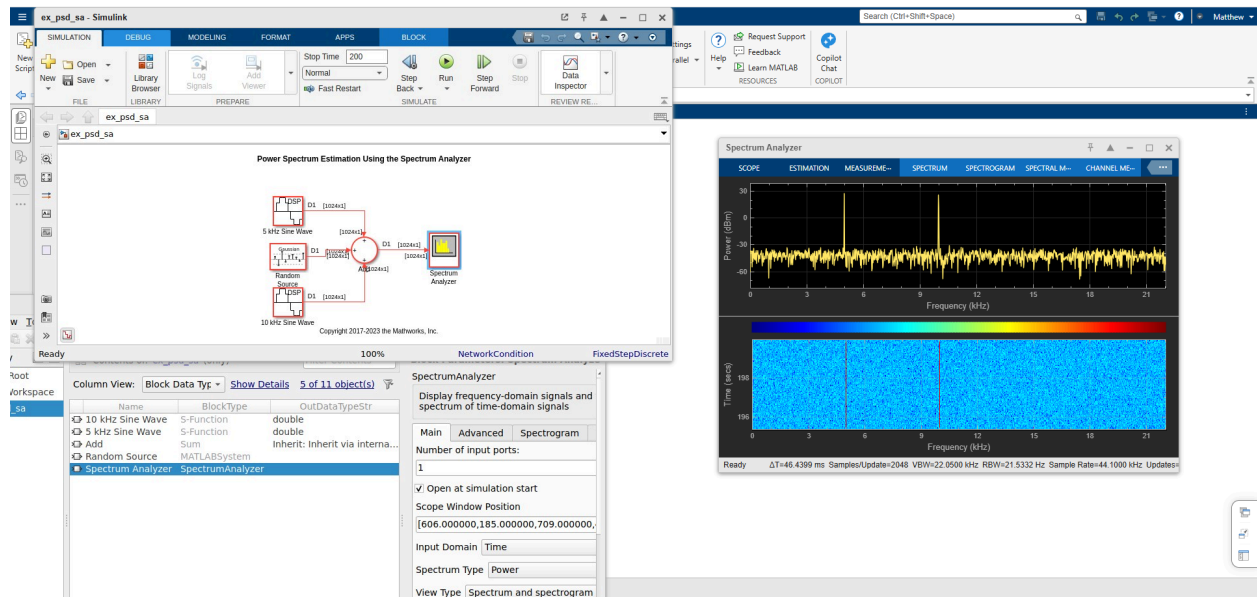


MATLAB Interlude 2

Initial Version Capture:



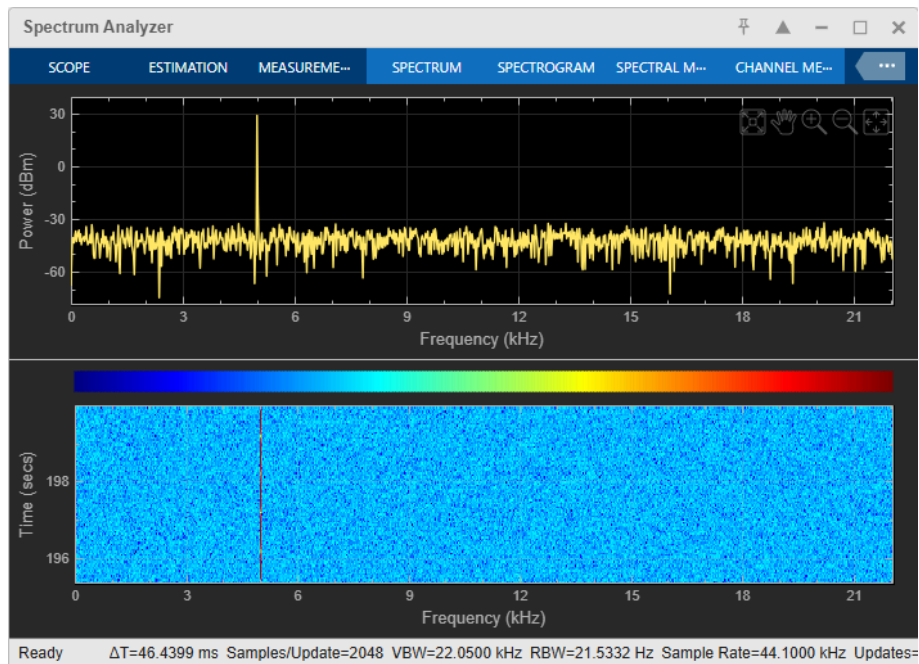
Resolution Bandwidth (RBW)

Resolution Bandwidth is a frequency bandwidth that resolves closely spaced spectral components that would otherwise be unseen under a wider frequency range. A narrow RBW allows an engineer to locate a signal close in frequency whereas a larger RBW may blur them together. The benefits of a narrow RBW allows for more precise measurements at the cost of efficiency. A wider RBW allows for quicker analysis at the cost of precision. That is why areas of engineering such as telecommunications and aerospace require narrow RBWs. Essentially RBW is an intermediate filter that allows a certain bandwidth of frequencies to pass according to an engineer's design or testing specification.

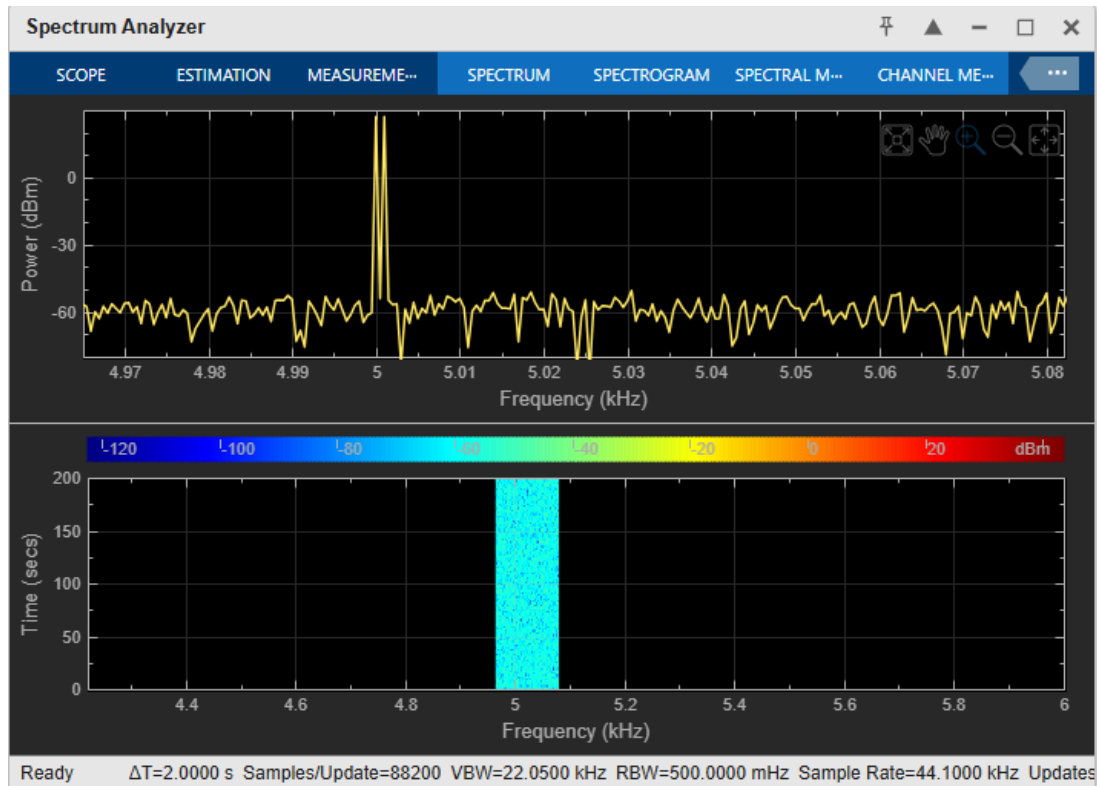
Link used for research:

<https://www.keysight.com/used/us/en/knowledge/glossary/oscilloscopes/what-is-resolution-bandwidth>

Altered frequency capture:

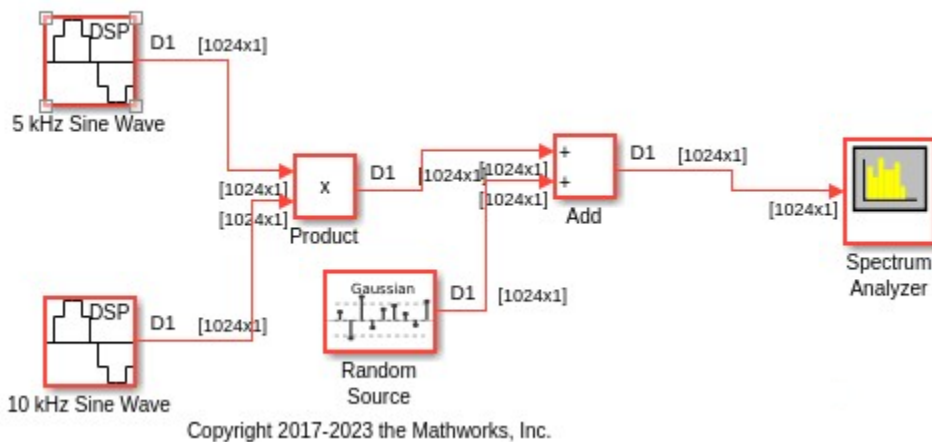
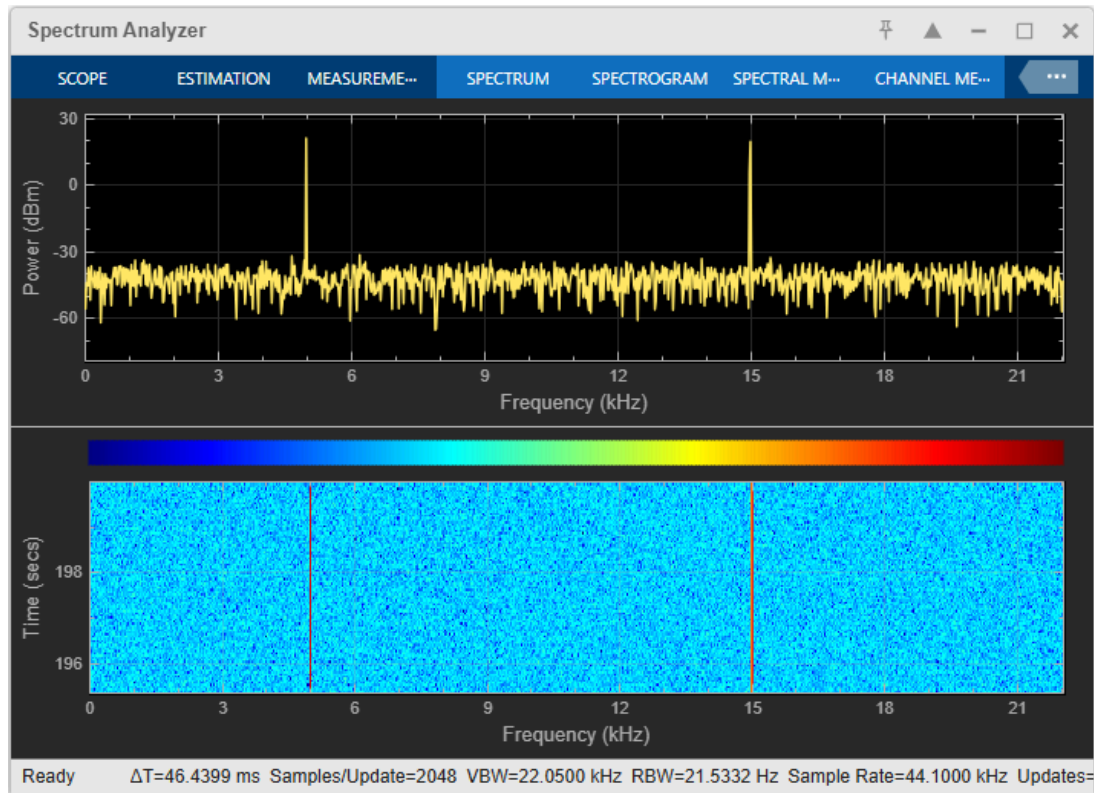


Frequencies selected: 5000 hz and 5001 hz



RBW selection: 0.5 Hz

Non-Linear System



In order to make this system non-linear I added a product block and rearranged the existing blocks to create the product of the two signals with some simulated Gaussian white noise. The product of both signals displays frequency spikes at 5 khz and 15 khz accordingly.